

Part G: Nmap Scan Report

Scan Report

Scan Date: 28 June 2026

Target System: localhost (127.0.0.1)

Commands Used

```
nmap --version  
nmap localhost  
nmap -sV localhost  
nmap -O localhost
```

Open Ports

Port	State
80/tcp	Open
135/tcp	Open
445/tcp	Open

Running Services

Port	Service	Version
80/tcp	HTTP	Microsoft IIS httpd 10.0
135/tcp	MSRPC	Microsoft Windows RPC
445/tcp	Microsoft-DS (SMB)	Version Not Detected

Operating System

- **Operating System:** Microsoft Windows 11 (24H2 – 25H2)

Observations

- The target system was reachable and responded successfully to all Nmap scans.

- Three TCP ports (80, 135, and 445) were found to be open.
- The HTTP service was running on Microsoft IIS 10.0.
- Microsoft Windows RPC and SMB services were also active.
- Nmap successfully detected the operating system as Microsoft Windows 11.

Recommendations

1. Disable unnecessary services and close unused ports.
2. Keep Windows and all services updated with the latest security patches.
3. Restrict access to SMB (Port 445) and RPC (Port 135) using a firewall.
4. Regularly monitor open ports and network services.
5. Use strong authentication and security configurations for all running services.

Conclusion

During this practical, I learned how to use Nmap to perform basic network reconnaissance and analyze a local system. I verified the installation of Nmap, scanned my computer to identify open ports, detected running services, and performed operating system detection. The scan results showed that ports **80 (HTTP)**, **135 (MSRPC)**, and **445 (SMB)** were open. Nmap also identified the operating system as **Microsoft Windows 11 (24H2–25H2)**. By using the **-sV** option, I was able to detect service versions, including Microsoft IIS 10.0 running on the HTTP service. This activity demonstrated how network scanning helps identify active services and potential attack surfaces on a system. I also learned that unnecessary open ports and improperly configured services can increase security risks. Regular scanning, timely software updates, proper firewall configuration, and disabling unused services are important practices for improving system security. Overall, this exercise enhanced my understanding of reconnaissance techniques, port scanning, service identification, and operating system detection, which are essential skills in cybersecurity and penetration testing.